Ankita Negi

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Researcher in Physics and Data Analysis

I hold a Dr. rer. nat. in Physics with strong expertise in data analysis, algorithm development and advanced problem solving. I am proficient in Python programming, machine learning tools and data management. I have experience in inverse problems, mathematical optimization, digital signal processing and computational imaging methods - specifically ptychography. During my research career, I thrived in both independent and collaborative settings and have worked in international, interdisciplinary, multi-team projects. I come from a synchrotron, optics and condensed matter science background, and am highly motivated to apply my modeling and analytical skills to real world technologies.

Skills

Programming: Python (NumPy, Pandas, SciPy, scikit-learn, PyTorch, Polars), SQL, MATLAB, Mathematica, Julia
Data Science: Supervised/unsupervised machine learning, regression, classification, statistical modeling
Tools: Git, pytest, Linux, Conda/Mamba/Pixi, SLURM
Signal Processing Inverse problems, regularization, simulation, noise modeling, numerical optimization, differential equations, computational imaging algorithms (ptychography, crystallography, tomography, holography, magnetic particle imaging)
Data Visualization: Matplotlib, PyVista, Mayavi
Languages: English (C1), German (B1), Hindi (Native)

Experience

Center for Ultrafast Imaging, Postdoctoral Researcher — Hamburg, Germany, Jun 2024 – Jan 2026

- Worked on large-scale scientific computing for X-ray and optical imaging experiments in collaboration with ESRF and DESY.
- Contributed to development and coordination of experimental pipelines for advanced detector systems.
- Designed and supported data analysis workflows in an international research environment.

Deutsches Elektronen-Synchrotron (DESY), Doctoral Researcher — Hamburg, Germany, Oct 2020 – Apr 2024

- Conducted synchrotron-based research in materials spectroscopy and inverse problems.
- Planned and led beamtime experiments at ESRF and PETRA III facilities.
- Collaborated in international research projects and presented results at scientific conferences.

University of Cologne, Student Assistant — Cologne, Germany, Oct 2018 – Jan 2019

- Supported computational physics research in condensed matter theory.
- Performed numerical simulations on HPC clusters and assisted in teaching activities.

IISER Mohali, Research Intern — Mohali, India, Jun 2017 – Aug 2017

- Conducted computational simulations for astrophysics research problems.

Education

Ph.D. **University of Hamburg**, Physics - CGPA: 1.0, *magna cum laude* — Oct 2020 – April 2024

- Member of the Helmholtz Data Science Graduate School (DASHH).
- Advanced ptychography and quantitative imaging algorithms,

M.Sc. **University of Cologne**, Physics - CGPA: 1.4 — Oct 2017 – Jan 2019

- Member of Collaborative Research Center (CRC) 183 -Entangled States of matter.
- Specialized in computational condensed matter physics and biophysics.

B.Sc. (Hon.) **University of Delhi**, Physics - Grade: 94.9% — 2014 – 2017

- Executive Council Member, Physics and Electronics Society, St. Stephen's College.

Selected Publications

- Interferometric measurement of nuclear resonant phase shift with a nanoscale Young double waveguide, Nature Photonics (2026), *Co-author*
- Energy Time Ptychography for one-dimensional phase retrieval, Optica (2025), *First author*
- *Collective Nuclear Excitation and Pulse Propagation in Single-Mode X-Ray Waveguides*, Physical Review Letters (2025), *Co-author*

Awards

- 3rd place, Helmholtz Data Science Challenge - Machine learning on real world datasets — 2022
- Bonn-Cologne Graduate School Scholarship — 2018-2019
- IAS–INSA–NASI Summer Research Fellowship — 2017
- INSPIRE fellowship, Department of Science and Technology, Government of India — 2014-2017

Projects

Scientific Software Development

- Developed NuPty, an open-source Python/PyTorch framework for inverse problems and reconstruction of material properties from noisy, high-dimensional experimental data.
- Built modular Python libraries for calibration, fitting, and optimization workflows in X-ray imaging experiments, including calibration tools and modeling routines. These tools are being used in multiple academic research projects (1 Bachelor thesis, 2 Master's theses, 1 PhD project).

Computational Imaging & Reconstruction

- Developed a reconstruction framework for nuclear ptychography and inverse problems in synchrotron spectroscopy, enabling high-resolution analysis of material structure from experimental scattering data.
- Developed a machine learning pipeline for microscopy image classification using CycleGAN-based augmentation and PCA-based features, improving accuracy by **20%** and achieving **3rd place** in a data challenge. Project repository

High-Performance Computing & Data Processing

- Processed multi-gigabyte experimental datasets using HPC clusters, implementing scalable data reduction and reconstruction workflows.
- Performed large-scale numerical simulations of Hamiltonian systems using transfer matrix methods, including parameter sweeps and phase diagram construction.

Conferences, Workshops and Seminars

2026

- High Performance Computing with Python, Jülich Supercomputing Centre, 2026.
- Multivariate Statistics Workshop, HIDA, 2026.

2025

- Regularization in Image Reconstruction Workshop, HIDA, 2025.
- Image Processing Lectures: Six Main Tasks, HIDA, 2025.
- Oral presentation, DPG Spring Meeting, Bonn, 2025.

2024–2023

- Deep Learning in Inverse Problems Workshop, Hamburg, 2024.
- Phase Retrieval and Coherent Scattering Conference (Coherence), Sweden, 2024.
- Ptychography Software Workshop, Diamond Light Source, UK, 2023.
- Oral presentations at DESY Photon Science Users' Meeting and international conferences, 2023–2024.